

Paper 1 - Set 03 - Solutions

Generated practice paper: answer key and worked solutions.

Answer key

Question	Answer	Topic
1	A	Statistics
2	C	Statistics
3	B	Polynomials, factor and remainder theorems
4	D	Indices and surds
5	A	Graphs of functions
6	A	Straight lines
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10	A	Sequences and series
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19	A	Triangles, radians and sectors
20	B	Differentiation

Worked solutions

Question 1 STA-01 · Statistics · spec M6.3

The mean of five numbers is 12. When a sixth number is added the mean becomes 14. What is the sixth number?

Answer: A

Total goes from 60 to 84: the sixth number is 24.

Full method - how and why

Means questions are almost always really about **totals**. Convert each mean into a total before doing anything else.

1 Original total $= 5 \times 12 = 60$.

why: $\text{mean} = \frac{\text{total}}{\text{how many}}$, so $\text{total} = \text{mean} \times \text{how many}$.

2 New total $= 6 \times 14 = 84$.

why: There are now six numbers, and their mean is 14.

3 The sixth number is the difference: $84 - 60 = 24$.

why: Nothing else changed, so the whole increase in the total is the new number.

Where it goes wrong

- Answering 14 - the new **mean**, not the new **number**.
- Answering 84, the new total rather than the added value.
- Thinking the new number must be 2 more than the old mean; the mean rose by 2 for **all six**, so the newcomer must carry $6 \times 2 = 12$ extra above the old mean of 12, giving 24.

Takeaway Turn means into totals; the added value is the change in total.

Question 2 STA-03 · Statistics · spec M6.3

The numbers 3, 7, x , 12, 18 (with $7 \leq x \leq 12$) have mean equal to median. Find x .

Answer: C

The median is x and the mean is $\frac{40+x}{5}$. Setting them equal: $40 + x = 5x$, so $x = 10$.

Full method - how and why

Write expressions for the mean and the median, then equate. The given range for x tells you where it sits in the order.

1 Since $7 \leq x \leq 12$, the sorted list is 3, 7, x , 12, 18, so the median is x .

why: With five values the median is the third; the constraint is there precisely to fix x 's position.

2 Mean $= \frac{3 + 7 + x + 12 + 18}{5} = \frac{40 + x}{5}$.

why: Sum the known values first: $3 + 7 + 12 + 18 = 40$.

3 Set mean = median: $\frac{40 + x}{5} = x$.

why: That is the condition stated.

4 Solve: $40 + x = 5x \Rightarrow 4x = 40 \Rightarrow x = 10$.

why: And 10 does lie in $[7, 12]$, so the assumed ordering holds.

Where it goes wrong

- Taking the median to be 12 (the middle of the numbers as originally listed without x).
- Forgetting to check that the answer is consistent with the assumed position in the order.

Takeaway Order first, then write the median; always check the solution respects the ordering you assumed.

Question 3 POL-03 · Polynomials, factor and remainder theorems · spec MM1.6

Fully factorise $x^3 - 7x + 6$.

Answer: B

$f(1) = 0$, so $(x - 1)$ is a factor: $x^3 - 7x + 6 = (x - 1)(x^2 + x - 6) = (x - 1)(x + 3)(x - 2)$.

Full method - how and why

Factorising a cubic: find one root by inspection, divide it out, then factorise the quadratic that remains.

- 1 Test small factors of the constant term 6: $f(1) = 1 - 7 + 6 = 0$, so $(x - 1)$ is a factor.
why: Any integer root must divide the constant term, so only $\pm 1, \pm 2, \pm 3, \pm 6$ are worth testing - and 1 is always the quickest to try.
- 2 Divide: $x^3 - 7x + 6 = (x - 1)(x^2 + x - 6)$.
why: Note the missing x^2 term in the original - treat it as $0x^2$ when dividing or comparing coefficients.
- 3 Factorise the quadratic: $x^2 + x - 6 = (x + 3)(x - 2)$.
why: Two numbers multiplying to -6 and adding to $+1$.
- 4 Full factorisation: $(x - 1)(x - 2)(x + 3)$.
why: Check by multiplying the constants: $(-1)(-2)(3) = 6$. ✓

Where it goes wrong

- Forgetting the missing x^2 term and producing the wrong quadratic factor.
- Sign errors giving $(x - 1)(x + 2)(x - 3)$, whose constant term is $+6$ too - always check the x coefficient as well.

Takeaway Root-hunt among the divisors of the constant term, divide out, then factorise what remains; check by multiplying the constants back.

Question 4 IND-20 · Indices and surds · spec MM1.2

Given that $\sqrt{a} - \sqrt{b} = 2$ and $a - b = 12$, find $\sqrt{a} + \sqrt{b}$.

Answer: D

$a - b = (\sqrt{a} - \sqrt{b})(\sqrt{a} + \sqrt{b})$, so $12 = 2(\sqrt{a} + \sqrt{b})$, giving 6.

Full method - how and why

You are given a difference of surds and a difference of the numbers. The link is the difference of two squares - no need to find a and b .

- 1 Factorise $a - b$ as a difference of two squares in $\sqrt{a}$ and $\sqrt{b}$: $a - b = (\sqrt{a} - \sqrt{b})(\sqrt{a} + \sqrt{b})$.
why: $a = (\sqrt{a})^2$ for $a \geq 0$, so the whole expression is $X^2 - Y^2$ with $X = \sqrt{a}$, $Y = \sqrt{b}$ - this is the only structural idea in the question.
- 2 Substitute the given values: $12 = 2 \times (\sqrt{a} + \sqrt{b})$.
why: The first bracket is exactly the quantity given as 2.
- 3 Divide: $\sqrt{a} + \sqrt{b} = 6$.
why: Direct.

Where it goes wrong

- Solving simultaneously for a and b (possible: $a = 16, b = 4$) - correct but three times the work and easy to fumble.
- Answering 24 by multiplying rather than dividing.

Takeaway $a - b = (\sqrt{a} - \sqrt{b})(\sqrt{a} + \sqrt{b})$ converts between sums and differences of surds instantly.

Question 5 GRA-06 · Graphs of functions · spec MM8.6

The curve $y = x^4 - 5x^2 + 4$ meets the coordinate axes at how many points in total?

Answer: A

x -axis: $(x^2 - 1)(x^2 - 4) = 0$, four points. y -axis: $(0, 4)$, one point. Total 5.

Full method - how and why

Count both axes. The x -axis meetings come from the roots; the y -axis always contributes exactly one point.

1 Factorise: $x^4 - 5x^2 + 4 = (x^2 - 1)(x^2 - 4)$.

why: Quadratic in x^2 : two numbers multiplying to 4 and adding to -5 .

2 Solve: $x = \pm 1$ and $x = \pm 2$ - four x -axis crossings.

why: Each positive value of x^2 gives two values of x .

3 y -axis: substitute $x = 0$ to get $y = 4$ - one point.

why: A function meets the y -axis exactly once.

4 Total: $4 + 1 = 5$.

why: The y -intercept is a separate point (it is not one of the roots here, since $y(0) \neq 0$).

Where it goes wrong

- Forgetting the y -axis and answering 4.
- Counting $x^2 = 1, 4$ as only two points.

Takeaway 'Meets the coordinate axes' means both axes: roots plus the single y -intercept.

Question 6 LIN-05 · Straight lines · spec MM3.1

Which of the following lines is parallel to $4x + 6y = 9$?

Answer: A

$4x + 6y = 9$ has gradient $-\frac{2}{3}$; so does $2x + 3y = 1$.

Full method - how and why

Parallel means equal gradient. Convert each option to gradient form - or compare the coefficient ratios directly.

1 Gradient of $4x + 6y = 9$ is $-\frac{4}{6} = -\frac{2}{3}$.

why: For $ax + by = c$ the gradient is $-\frac{a}{b}$.

2 Test the options: $2x + 3y = 1$ has gradient $-\frac{2}{3}$ ✓.

why: Equivalently, the coefficients $(2, 3)$ are proportional to $(4, 6)$ - the constant term is irrelevant to direction.

3 Reject the rest: $3x - 2y = 4$ gives $\frac{3}{2}$ (perpendicular), $4x - 6y = 9$ gives $\frac{2}{3}$, $6x + 4y = 9$ gives $-\frac{3}{2}$, and $y = \frac{3}{2}x + 1$ gives $\frac{3}{2}$.

why: Note how many distractors are the **perpendicular** gradient - a deliberate trap.

Where it goes wrong

- Choosing $6x + 4y = 9$ because the numbers 4 and 6 reappear - but swapped, so the gradient changes.
- Forgetting the minus sign in $-\frac{a}{b}$.

Takeaway Parallel $\Leftrightarrow$ the coefficients of x and y are in the same ratio; the constant plays no part.

Question 7 INT-08 · Integration · spec MM7.4

Let $F(x) = \int_2^x (t^2 - 3t) dt$. Find $F'(3)$.

Answer: E

By the Fundamental Theorem of Calculus $F'(x) = x^2 - 3x$, so $F'(3) = 9 - 9 = 0$.

Full method - how and why

A direct test of the Fundamental Theorem of Calculus: differentiating an integral with a variable upper limit gives back the integrand.

- 1 Apply the FTC: $\frac{d}{dx} \int_2^x g(t) dt = g(x)$, so $F'(x) = x^2 - 3x$.
why: The lower limit is a constant, so it contributes nothing to the derivative.
- 2 Substitute $x = 3$: $F'(3) = 9 - 9 = 0$.
why: Just evaluate the integrand at 3 - no integration is needed at all.

Where it goes wrong

- Actually integrating and then differentiating - correct but slow, and error-prone.
- Substituting $t = 3$ into an antiderivative and answering $-\frac{7}{6}$ or $\frac{9}{2}$, both offered.

Takeaway $\frac{d}{dx} \int_a^x f(t) dt = f(x)$: differentiation undoes integration, so just evaluate the integrand.

Question 8 TRG-07 · Trigonometric functions and equations · spec MM4.6

Solve $\sin^2 x = \frac{3}{4}$ for $0 \leq x < 2\pi$. How many solutions are there?

Answer: E

$\sin x = \pm \frac{\sqrt{3}}{2}$: $x = \frac{\pi}{3}, \frac{2\pi}{3}, \frac{4\pi}{3}, \frac{5\pi}{3}$. Four solutions.

Full method - how and why

A squared trig equation. Taking the square root produces **two** values, each with its own solutions.

- 1 Take square roots: $\sin x = \pm \frac{\sqrt{3}}{2}$.
why: $X^2 = k$ has two roots - forgetting the negative one halves the answer.
- 2 $\sin x = \frac{\sqrt{3}}{2}$ gives $x = \frac{\pi}{3}$ and $\frac{2\pi}{3}$.
why: Second-quadrant partner: $\pi - \frac{\pi}{3}$.
- 3 $\sin x = -\frac{\sqrt{3}}{2}$ gives $x = \frac{4\pi}{3}$ and $\frac{5\pi}{3}$.
why: Third and fourth quadrants, by symmetry below the axis.
- 4 Total: 4 solutions in $[0, 2\pi)$.
why: Count them; symmetry says there should be an even number here.

Where it goes wrong

- Taking only the positive root and answering 2.
- Reading $\sin^2 x$ as $\sin(x^2)$.

Takeaway $\sin^2 x = c$ gives $\sin x = \pm \sqrt{c}$, and each sign contributes its own solutions.

Question 9 INT-16 · Integration · spec MM7.4

Given that $\int_0^4 f(x) dx = 10$, find $\int_0^4 (f(x) + 3x) dx - \int_4^0 f(x) dx$.

Answer: D

$$10 + 24 - (-10) = 44.$$

Full method - how and why

Three integral rules in one: linearity, the integral of $3x$, and the effect of swapping limits.

1 Split the first integral: $\int_0^4 (f(x) + 3x) dx = \int_0^4 f(x) dx + \int_0^4 3x dx = 10 + \int_0^4 3x dx$.

why: Sums split into separate integrals.

2 Compute $\int_0^4 3x dx = \left[\frac{3x^2}{2} \right]_0^4 = 24$.

why: $\frac{3 \times 16}{2} = 24$.

3 Handle the second term: $\int_4^0 f(x) dx = -\int_0^4 f(x) dx = -10$.

why: Reversing the limits reverses the sign - the key fact here.

4 Combine: $(10 + 24) - (-10) = 44$.

why: Subtracting a negative adds.

Where it goes wrong

- Treating $\int_4^0 f$ as $+10$ and answering 24.
- Forgetting the $3x$ term, giving 20.

Takeaway $\int_b^a = -\int_a^b$; watch for reversed limits used as a deliberate trap.

Question 10 SEQ-16 · Sequences and series · spec MM2.2

The sum of the first n terms of an arithmetic series is S_n . Given that $S_{10} = 145$ and $S_{20} = 490$, find the common difference.

Answer: A

$$S_{10} = 5(2a + 9d) = 145 \Rightarrow 2a + 9d = 29; S_{20} = 10(2a + 19d) = 490 \Rightarrow 2a + 19d = 49. \text{ Subtract: } 10d = 20, d = 2.$$

Full method - how and why

Two sums give two equations in a and d . Subtracting eliminates a - and only d is wanted.

1 $S_{10} = \frac{10}{2}[2a + 9d] = 145$, so $2a + 9d = 29$.

why: Divide out the $\frac{n}{2}$ factor straight away to keep the equations clean.

2 $S_{20} = \frac{20}{2}[2a + 19d] = 490$, so $2a + 19d = 49$.

why: Same formula with $n = 20$.

3 Subtract: $10d = 20$, so $d = 2$.

why: The $2a$ terms are identical in both equations, so subtraction removes them at once.

Where it goes wrong

- Using $(n - 1)d$ as $10d$ and $20d$ rather than $9d$ and $19d$.
- Dividing 490 by 20 instead of by 10.

Takeaway Two sums, two unknowns: write both in the $2a + (n - 1)d$ form and subtract to eliminate a .

Question 11 BIN-13 · Binomial expansion · spec MM2.4

Use the binomial expansion to find the value of 1.01^5 correct to 4 decimal places.

Answer: B

$$(1 + 0.01)^5 = 1 + 5(0.01) + 10(0.0001) + 10(0.000001) + \dots = 1 + 0.05 + 0.001 + 0.00001 + \dots \approx 1.0510.$$

Full method - how and why

An approximation with a small quantity. The binomial expansion converges fast because 0.01 is tiny.

1 Write $1.01^5 = (1 + 0.01)^5$ and expand: $1 + 5(0.01) + 10(0.01)^2 + 10(0.01)^3 + \dots$

why: Coefficients from row 5 of Pascal's triangle: 1, 5, 10, 10, 5, 1.

2 Evaluate the terms: $1 + 0.05 + 0.001 + 0.00001 + \dots$

why: $(0.01)^2 = 0.0001$, so the third term is $10 \times 0.0001 = 0.001$.

3 Add as far as the 4-decimal-place accuracy requires: $1.05101\dots \approx 1.0510$.

why: The fourth term only affects the fifth decimal place, so three terms suffice.

Where it goes wrong

- Stopping after two terms and answering 1.0500.
- Mis-scaling $(0.01)^2$ as 0.001, which gives 1.0600-style errors.

Takeaway With a small x , each binomial term is much smaller than the last - take just enough terms for the required accuracy.

Question 12 DIF-09 · Differentiation · spec MM6.3

A rectangle has perimeter 40. What is its maximum possible area?

Answer: E

Sides x and $20 - x$: $A = 20x - x^2$, $A' = 20 - 2x = 0$ at $x = 10$, giving $A = 100$ (a square).

Full method - how and why

A standard optimisation: express the area in one variable using the constraint, then differentiate.

1 Let one side be x ; the perimeter condition $2x + 2y = 40$ gives $y = 20 - x$.

why: Use the constraint to eliminate the second variable - optimisation needs a single variable.

2 Write the area: $A = x(20 - x) = 20x - x^2$.

why: A quadratic in x .

3 Differentiate and solve: $A' = 20 - 2x = 0$ at $x = 10$.

why: Or complete the square: $A = 100 - (x - 10)^2$, which shows the maximum immediately.

4 Compute the area: $A = 10 \times 10 = 100$ (a square).

why: $A'' = -2 < 0$ confirms a maximum.

Where it goes wrong

- Using $x + y = 40$ instead of $2x + 2y = 40$, giving 400 - an offered option.
- Reporting the side length 10 rather than the area.

Takeaway For fixed perimeter, the rectangle of greatest area is the square - but show it via the constraint and the derivative.

Question 13 IND-10 · Indices and surds · spec MM1.1

The value of $\frac{2^{n+2} + 2^n}{2^{n+1}}$ is

Answer: E

Factor 2^n : numerator $= 2^n(4 + 1) = 5 \cdot 2^n$; denominator $= 2 \cdot 2^n$. Quotient $= \frac{5}{2}$.

Full method - how and why

Powers with an unknown exponent. Factorising out the smallest power turns the expression into a number.

- 1 Write every term with the common factor 2^n : $2^{n+2} = 4 \cdot 2^n$, $2^{n+1} = 2 \cdot 2^n$.

why: $2^{n+k} = 2^k \cdot 2^n$. Choosing the *smallest* power (2^n) as the common factor makes every remaining factor a whole number.

- 2 Substitute: $\frac{4 \cdot 2^n + 2^n}{2 \cdot 2^n} = \frac{5 \cdot 2^n}{2 \cdot 2^n}$.

why: Factorising the numerator is legitimate because both terms contain 2^n .

- 3 Cancel 2^n : the value is $\frac{5}{2}$, independent of n .

why: $2^n \neq 0$ for every n , so cancelling is always valid here.

Where it goes wrong

- 'Cancelling' the n 's in the indices, e.g. claiming $\frac{2^{n+2}}{2^{n+1}} = 2^2/2^1$ term by term across a sum - you cannot split a fraction over a sum in the denominator.
- Answering 2^n because the n 'has to appear somewhere'.

Takeaway With unknown exponents, factor out the lowest power; the n almost always cancels and the answer is a plain number.

Question 14 ALG-13 · GCSE algebra · spec M4.5

Factorise $6x^2 - 7x - 20$.

Answer: C

$$(3x + 4)(2x - 5) = 6x^2 - 15x + 8x - 20 = 6x^2 - 7x - 20.$$

Full method - how and why

Factorising $ax^2 + bx + c$ with $a \neq 1$. Use the 'split the middle term' method, or expand the options.

- 1 Multiply a and c : $6 \times (-20) = -120$. Find two numbers with product -120 and sum -7 : -15 and 8 .

why: This is the standard method; the pair always exists when the quadratic factorises over the integers.

- 2 Split the middle term: $6x^2 - 15x + 8x - 20$.

why: Order does not matter.

- 3 Factor in pairs: $3x(2x - 5) + 4(2x - 5) = (3x + 4)(2x - 5)$.

why: The bracket must match in both halves - that is the check that the split was right.

- 4 Verify by expanding: $6x^2 - 15x + 8x - 20 = 6x^2 - 7x - 20$ ✓.

why: Expanding the chosen option is the fastest check in a multiple-choice setting.

Where it goes wrong

- Choosing $(3x - 5)(2x + 4)$, which has a common factor 2 in the second bracket - a sign it cannot be right for this expression.
- Getting the signs the wrong way round: $(3x - 4)(2x + 5)$ gives $+7x$.

Takeaway Split the middle term using two numbers with product ac and sum b ; then expand your answer to check.

Question 15 PRO-05 · Probability · spec M7.2

A spinner has probability 0.15 of landing on red. It is spun 200 times. What is the expected number of times it lands on red?

Answer: A

$$0.15 \times 200 = 30.$$

Full method - how and why

Expected number of successes in repeated independent trials is simply $n \times p$.

- 1 Identify $n = 200$ spins and $p = 0.15$.

why: Each spin is an independent trial with the same probability.

- 2 Expected number $= 200 \times 0.15 = 30$.

why: $0.15 = \frac{15}{100}$, and 15% of 200 is 30.

Where it goes wrong

- Answering 15 by reading the percentage as the answer.
- Expecting the answer to be a whole number in general - expectations often are not, and that is fine.

Takeaway Expected frequency $= n \times p$.

Question 16 BIN-04 · Binomial expansion · spec MM2.4

What is the sum of the coefficients in the expansion of $(3x - 2)^7$?

Answer: D

Put $x = 1$: $(3 - 2)^7 = 1$.

Full method - how and why

Sum of all coefficients: substitute $x = 1$. No expansion needed at all.

- 1 Note that setting $x = 1$ makes every power of x equal to 1, so the expansion collapses to the sum of its coefficients.

why: If $(3x - 2)^7 = \sum c_k x^k$ then putting $x = 1$ gives $\sum c_k$.

- 2 Evaluate the original expression at $x = 1$: $(3 - 2)^7 = 1^7 = 1$.

why: You never need a single coefficient.

Where it goes wrong

- Expanding and trying to add eight terms under time pressure.
- Substituting $x = 0$, which gives only the constant term $(-2)^7 = -128$ - an offered option.

Takeaway $f(1)$ is the sum of the coefficients; $f(0)$ is the constant term; $f(-1)$ is the alternating sum.

Question 17 FUN-04 · Functions and transformations · spec MM1.7

Which of the following functions, each with domain the set of all real numbers, is one-to-one?

Answer: D

$x^3 + x$ is strictly increasing (its derivative $3x^2 + 1 > 0$), so different inputs give different outputs. $x^3 - x$ takes the value 0 at $x = -1, 0, 1$; the others are even or periodic.

Full method - how and why

'One-to-one' means no horizontal line meets the graph twice. The quickest test is whether the function is always increasing (or always decreasing).

- 1 Rule out the even and periodic functions: $|x|$ and $x^2 + 2x$ are symmetric (e.g. $|1| = |-1|$), and $\cos x$ repeats every 2π .

why: Any symmetry or repetition immediately produces two inputs with the same output.

- 2 Test $x^3 - x$: it equals 0 at $x = -1, 0, 1$ - three inputs with the same output.

why: A cubic with three distinct roots is certainly not one-to-one.

- 3 Test $x^3 + x$: its derivative $3x^2 + 1$ is always positive, so the function is strictly increasing.

why: A strictly increasing function can never take the same value twice, which is exactly one-to-one.

- 4 So $f(x) = x^3 + x$ is the one-to-one function.

why: Alternatively, x^3 and x both increase, so their sum does.

Where it goes wrong

- Assuming every cubic is one-to-one - $x^3 - x$ is a counterexample.
- Thinking $|x|$ is one-to-one because it has a 'simple' formula.

Takeaway One-to-one $\Leftrightarrow$ strictly monotonic; check for symmetry or repeated values first, then look at the derivative's sign.

Question 18 SIM-10 · Simultaneous equations and inequalities · spec MM1.5

Find the complete set of values of x for which $(x - 1)(x - 3)(x - 5) > 0$.

Answer: A

Sign chart: the product is positive for $x > 5$, negative for $3 < x < 5$, positive for $1 < x < 3$, negative for $x < 1$.

Full method - how and why

A cubic inequality. With three distinct linear factors, use a sign chart - the sign alternates as you cross each root.

- 1 The roots are $x = 1, 3, 5$; mark them on a number line, splitting it into four regions.

why: A product of linear factors can only change sign at a root.

- 2 Test the far-right region, e.g. $x = 6$: $(5)(3)(1) > 0$.

why: One test value fixes the whole chart; the far right is easiest because every factor is positive.

- 3 Alternate leftwards: positive for $x > 5$, negative on $(3, 5)$, positive on $(1, 3)$, negative for $x < 1$.

why: Each simple (unrepeated) root flips the sign exactly once.

- 4 Collect the positive regions: $1 < x < 3$ or $x > 5$.

why: The answer is a union of two intervals.

Where it goes wrong

- Answering $x > 5$ only, forgetting the middle positive region.
- Assuming the expression is positive whenever x is bigger than all the roots - true here, but the alternation is what you must use.

Takeaway Sign chart: find the roots, test one region, then alternate; repeated factors are the only ones that do not flip the sign.

Question 19 TRI-10 · Triangles, radians and sectors · spec MM4.1

In triangle ABC , $\sin A : \sin B : \sin C = 3 : 5 : 7$. Find angle C .

Answer: A

By the sine rule the sides are in ratio $3 : 5 : 7$. $\cos C = \frac{9 + 25 - 49}{2 \cdot 3 \cdot 5} = -\frac{15}{30} = -\frac{1}{2}$, so $C = 120^\circ$.

Full method - how and why

A ratio of sines is a ratio of sides (sine rule). Then the cosine rule finds the angle.

- 1 By the sine rule, $a : b : c = \sin A : \sin B : \sin C = 3 : 5 : 7$.

why: $\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$ means sides are proportional to the sines of opposite angles.

- 2 Take the sides as $3k, 5k, 7k$; the k will cancel, so use $3, 5, 7$.

why: Only the shape matters for angles, so the scale factor is irrelevant.

- 3 Apply the cosine rule for the angle opposite 7: $\cos C = \frac{9 + 25 - 49}{2(3)(5)} = \frac{-15}{30} = -\frac{1}{2}$.

why: C is opposite the largest side, so expect the largest angle.

- 4 $\cos C = -\frac{1}{2}$ gives $C = 120^\circ$.

why: One of the exact values: $\cos 120^\circ = -\cos 60^\circ = -\frac{1}{2}$.

Where it goes wrong

- Treating $3 : 5 : 7$ as the angles themselves, giving 84° .
- Reading $\cos C = -\frac{1}{2}$ as 60° and ignoring the sign.

Takeaway Sines in ratio $\Rightarrow$ sides in the same ratio; then use the cosine rule with any convenient multiple.

Question 20 DIF-16 · Differentiation · spec MM6.3

The tangent to $y = x^3$ at the point (a, a^3) , where $a \neq 0$, meets the curve again at the point with x -coordinate

Answer: B

Tangent: $y = 3a^2x - 2a^3$. Solve $x^3 - 3a^2x + 2a^3 = 0$; $x = a$ is a double root, so $(x - a)^2(x + 2a) = 0$: the other intersection is $x = -2a$.

Full method - how and why

A tangent meeting the curve again. The tangency shows up as a repeated root, which lets you factorise the cubic without solving it.

- 1 Find the tangent at (a, a^3) : $y' = 3x^2$, so the gradient is $3a^2$ and the line is $y = 3a^2(x - a) + a^3 = 3a^2x - 2a^3$.

why: Point-gradient form, then simplify.

- 2 Set the tangent equal to the curve: $x^3 = 3a^2x - 2a^3$, i.e. $x^3 - 3a^2x + 2a^3 = 0$.

why: Intersections satisfy both equations.

- 3 $x = a$ is a **repeated** root (that is what tangency means), so factor out $(x - a)^2$: $x^3 - 3a^2x + 2a^3 = (x - a)^2(x + 2a)$.

why: Check by comparing the constant term: $(-a)^2(2a) = 2a^3 \checkmark$.

- 4 The remaining root is $x = -2a$.

why: That is where the tangent cuts the curve again.

Where it goes wrong

- Forgetting that tangency gives a **double** root and trying to factor out only one $(x - a)$.
- Sign error giving $x = 2a$.

Takeaway A tangent meets the curve in a repeated root; factor $(x - a)^2$ out and read off the third root.